nature communications

Article

https://doi.org/10.1038/s41467-026-68782-w

Spinal cord stimulation therapy for gait impairment in Parkinson's disease: a double-blinded, randomised feasibility trial with an open extension

Received: 1 May 2025

Accepted: 16 January 2026

Published online: 29 January 2026

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A trial of spinal cord stimulation (SCS) was performed in people with gait-impaired Parkinson's (ClinicalTrials.gov: NCT05110053). Fourteen patients underwent gait assessments and [^{18}F]-FDG and [^{18}F]-FEOBV PET at baseline, six, and twelve months after SCS. Twelve participants were randomised to six-month MicroBurst or sham, followed by six-month extension with stimulation. The primary outcomes were feasibility and safety, as captured by the trial process measures and nature and frequency of adverse events, respectively, and the Postural Instability and Gait Disorder (PIGD) score as a clinical outcome. Secondary outcomes included assessments of balance and gait at home and at visits, including the Lower Body and Gait (LBG) score, imaging, and patient-reported outcomes of changes in gait, balance and quality of life. Seventeen patients (12%) were eligible for enrolment. Recruitment was feasible (1.2 participants/month) and SCS was well-tolerated. At six months, MicroBurst did not significantly improve gait compared with sham, although bradykinesia/rigidity improved. At 12 months, LBG scores improved (−4.31 points, $p = 0.0012$) with bilateral decreased thalamic metabolism and decreased right anterior insula [^{18}F]-FEOBV uptake. The trial met its primary feasibility and safety endpoints by achieving recruitment targets and demonstrating that SCS was well-tolerated. However, it did not meet the primary clinical endpoint of a significant PIGD improvement at six months. Larger trials are warranted, as SCS may improve LBG and leg rigidity/bradykinesia, especially as time progresses. We reported data for power calculations and identified important risks for designing future trials.

Gait impairment in idiopathic Parkinson's disease (PD) is one of the disabling manifestations of the disease that are often refractory to available treatments[1,2]. In particular, freezing of gait (FoG)[3], which is characterised by a transient feeling of being nailed to the floor despite the intention to walk, reduces patients' independence and significantly increases the risk of falls, affecting quality of life and decreasing life expectancy of PD patients[3].

Spinal cord stimulation (SCS) is a treatment for intractable pain that delivers electrical stimulation to the dorsal column through a lead implanted in the epidural space and connected to a subcutaneously

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implanted pulse generator (IPG)[4]. Over the last decade, SCS has been proposed to improve gait features in PD patients, as case reports and series have noted the benefit of conventional square-wave SCS in PD patients with back pain and gait impairment, with SCS appearing to relieve both to a considerable degree[5,6]. Additionally, a few trials provided evidence to suggest square-wave SCS could be beneficial for gait in PD patients without back pain[7,8]. However, due to conflicting findings especially in the long-term treatment, questions regarding its actual efficacy have been raised and interest has somewhat tapered[9–11]. Recently, burst stimulation, a newer SCS stimulation paradigm with regular trains of pulses, has also been shown to improve gait and balance as well as mental status in a few PD patients[12]. Unlike traditional SCS, burst stimulation does not induce paraesthesias and could therefore not only be a superior therapeutic option compared to traditional square-wave SCS but could also enable true blinding in research trials[13]. Nevertheless, literature on SCS for gait problems in PD is still sparse and potential mechanisms of action remain unclear[10]. Importantly, no double-blinded clinical trials of burst SCS on gait impairment have been reported so far.

The Spinal Cord Stimulation Therapy for Patients with Parkinson's Disease and Gait Problems (STEP-PD) study was designed as a feasibility trial but is also a prospective double-blinded, randomised-controlled trial (RCT) of SCS delivered as burst stimulation for gait impairment in PD so far, followed by an open extension phase[13]. Furthermore, this trial of SCS included [^{18}F]fluorodeoxyglucose (FDG) and [^{18}F]fluoroethoxybenzovesamicol (FEOBV) PET imaging. STEP-PD aims to 1) evaluate the safety and tolerability of SCS in PD patients with FoG, 2) assess any changes in cerebral metabolic rate or in cholinergic nerve

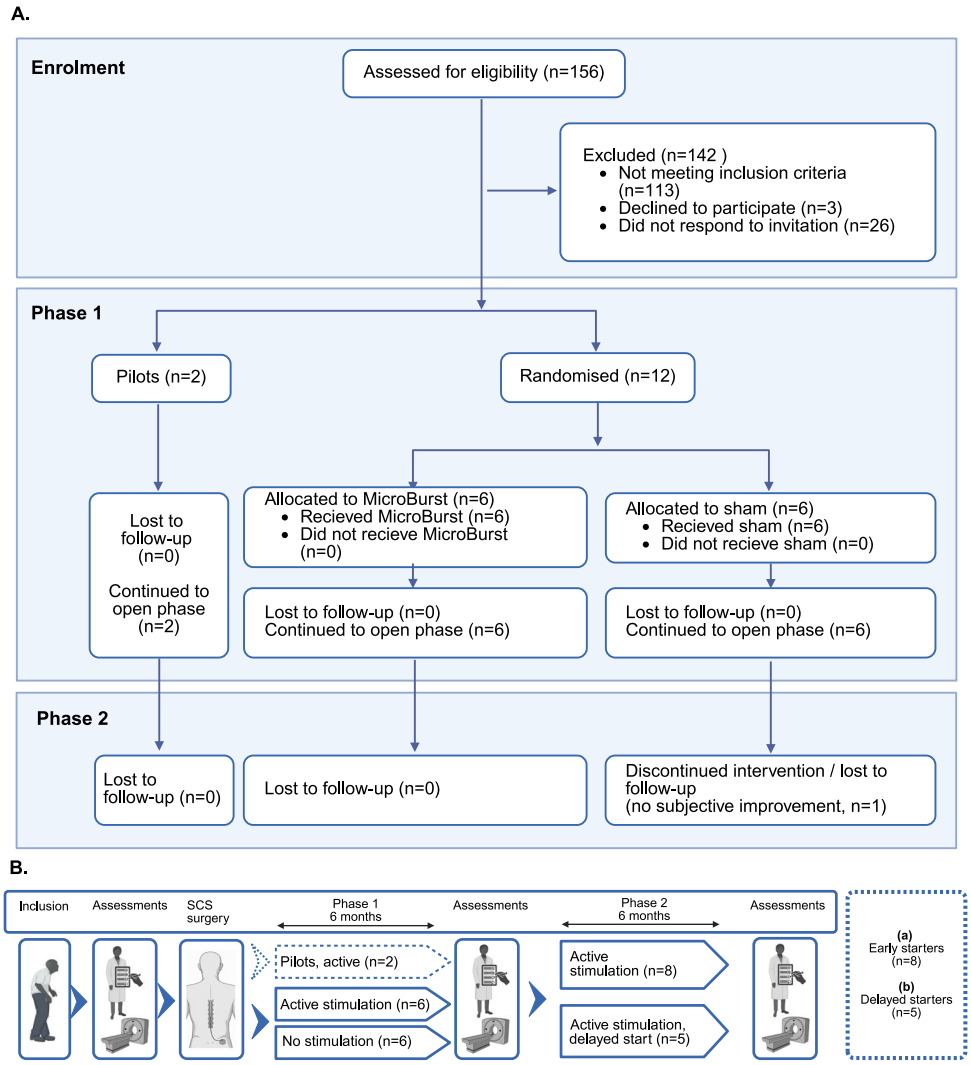

Fig. 1 | Participant enrolment and follow-up. A Recruitment and attrition: A total of 156 people was screened for enrolment, and 113 people did not meet the inclusion criteria as they did not have Parkinson's disease ($n = 16$), had back pain ($n = 46$), cognitive impairment ($n = 7$), prior deep brain stimulation ($n = 6$), other significant disabilities (significant other chronic pain, prior stroke, severe heart or lung disease, $n = 12$) or other reasons such as age below 50 years ($n = 3$), use of anticoagulants or antipsychotics ($n = 7$), not able to walk unaided for two minutes ($n = 6$), resident at nursing home ($n = 1$), newly started dopaminergic medicine or pump infusion ($n = 4$), no freezing of gait in on medication ON motor state ($n = 5$). During phase 1, all participants remained in their allocated group and proceeded to the open phase 2. During phase 2, one participant (originally allocated to the sham group with no stimulation) withdrew from the trial. **B** Study design: Following baseline assessments and spinal cord stimulation (SCS) surgery, participants either piloted the design or entered the randomised clinical trial (RCT) in phase 1. Subsequently, participants entered the open extension phase (phase 2), where all received active SCS. (a) Early starters ($n = 8$) include participants originally allocated to either pilots or active stimulation during phase 1. (b) Delayed starters include the participants originally allocated to sham and only turned on active stimulation in phase 2. Assessments at baseline and follow-ups comprised positron emission tomography imaging as well as the clinical assessments. In addition, patients wore a wearable device for gait assessment (STAT ON, Sense4Care, Barcelona, Spain). Created in BioRender. Terkelsen, M.H. (2025) https://BioRender.com/g1mda03.

terminal density related to SCS treatment, and 3) establish proof of concept and feasibility in terms of demand, adherence, recruitment and retention for a larger prospective multicentre study to assess efficacy of SCS in PD. Regarding the latter, we aimed to estimate standard deviations to enable sample size calculations for future trials and refine selection criteria.

Results

Recruitment, retention, and safety

Enrolment/participant flow and study design are presented in Fig. 1. For this single-centre trial, we had estimated a power of 96 % for the sample size of 12 participants to detect a three-point difference in mean changes from baseline in Postural Instability and Gait Disorder (PIGD) score, ($p < 0.05$) between groups[14,15]. A standard deviation (SD) of normal progression over 1 year in 709 patients with PD with gait problems (Movement Disorder Society – Unified Parkinson's Disease Rating Scale [MDS-UPDRS] Part III 3.10: Gait score ≥ 1) was obtained using the PPMI database[16]. One-hundred-forty patients with idiopathic PD were screened for enrolment, but only 12 % were eligible according to our criteria. Fourteen patients with idiopathic PD according to the MDS criteria[17] and FoG in the ON-state were enroled in the RCT over 12 months, corresponding to a recruitment rate of 1.2 participants per month. However, prior to surgery, one patient withdrew consent, and another patient was excluded as their baseline brain MRI showed the presence of a previous stroke (incidental finding; patient unaware). Two additional patients (one with back pain, exclusion criterion for the RCT) were enroled as pilot patients prior to the start of the RCT and underwent the same protocol of assessments, but stimulation was provided using an open-label design. The 12 patients who entered the RCT, yielding a consent rate of 80%, fulfilled all the criteria for participation and were randomised to either the MicroBurst or sham group. Their baseline characteristics and assessments are presented in Table 1. Individual data on all participants are detailed in Supplementary Tables 1–12. Briefly, demographic and clinical characteristics at baseline were similar between groups with some exceptions. The sham group consisted entirely of males at Hoehn and Yahr (H&Y) stage II; the MicroBurst group comprised four females and two males at H&Y stage II ($n = 3$) or III ($n = 3$). Compared with the sham group, the MicroBurst group had a higher (worse) Lower Body and Gait (LBG) score (17.00 ± 4.6 vs. 10.83 ± 2.5, $p = 0.017$) and higher (worse) sum of leg bradykinesia and rigidity items (12.17 ± 3.1 vs. 8.00 ± 0.9, p = 0.021) on the MDS-UPDRS[18]. Furthermore, they were slower (worse) on the three-metre timed-up-and-go (TUG) with cognitive Dual Task (DT) (19.99 ± 9.3 vs. 10.06 ± 1.4, $p = 0.048$) but experienced fewer freezing episodes per 1000 steps walked while wearing a wearable device (STAT-ON, Sense4Care, Barcelona, Spain)[19] (0.58 ± 0.5 vs. 1.54 ± 0.8, $p = 0.025$). However, the MicroBurst group did also have fewer steps per hour (576.02 ± 265.5 vs. 1059.73 ± 435.2, $p = 0.043$). There were no significant differences between groups for other measures, including the main outcome, the PIGD, although the MicroBurst group was in general slightly worse overall (Table 1, Supplementary Table 13).

There were no group differences in the timing of medication intake before assessments at baseline ($p = 0.987$) or follow-up ($p = 0.336$).

The randomised, double-blinded phase of STEP-PD was conducted between November 2021 and June 2023. All participants completed the trial period and adhered to the group assignment. The open extension phase of the STEP-PD trial ended in December 2023 (end-of-trial). During the open phase, attrition was 7% as one participant, originally allocated to the sham, decided to withdraw due to a lack of perceived effect. The two pilot participants completed the entire study period. Thus, at the end of the trial, the study population consisted of eight individuals who had received active stimulation for twelve months (early starters) and five individuals who switched from sham to active stimulation after the RCT (delayed starters). However, during the open extension phase, at the nine-month visit, some participants (original group allocation: pilots $n = 2$, sham $n = 2$, and MicroBurst $n = 2$) requested a change in stimulation paradigm to square-wave as any experienced improvement did not live up to their expectations. This deviation from our protocol was approved to maintain participants in the trial. Individual stimulation parameters are detailed in Supplementary Tables 14, 15.

Baseline assessments spanned 2–6 weeks due to scanner availability and patient scheduling constraints. All follow-up assessments were completed within two weeks, except one completed after 18 days.

Adverse events (AEs), stratified by trial group, are shown in Table 2. Four serious AEs were reported in the trial, and they were all related to the surgical implantation (dural puncture, $n = 1$; lead misplacement, $n = 2$; and perioperative pain, $n = 1$). Dural puncture and lead misplacement resulted in a second surgery with implantation of a lead placement, respectively, six weeks later and for that reason were classified as serious. In the case of perioperative pain, this resulted in the patient being admitted for one additional night as an inpatient. No further measures were required.

Postoperatively, all participants experienced some form of AE, most commonly transient and self-limited worsening in Parkinsonian symptoms (MicroBurst group: $n = 3$, sham group: $n = 6$, pilots: $n = 0$) for up to three months (one patient). This worsening of Parkinsonian symptoms did not require adjustment of dopaminergic medications. Second most common side effect was oedema at the site of implantation ($n = 5$). None required further intervention. No events requiring explanation occurred.

Three patients (original allocation: one MicroBurst, one sham, and one pilot) suffered falls requiring hospital contact during the study, as detailed in Table 2. All three patients had a history of falls requiring hospital contact prior to inclusion. No participants reported stimulation-induced side effects such as paraesthesias, dysesthesias, numbness or similar symptoms. No participants incurred adverse events related to device failure.

Clinical outcomes

End-of-treatment clinical outcomes at six-month follow-up are tabulated by group in Table 3. Regarding the main outcome, the PIGD, the MicroBurst group scored 0.69 (−3.23;1.85) points lower (better) compared with the sham group, although not statistically significant, $p = 0.556$. Regarding the LBG score, the MicroBurst group scored 2.9 (−8.3;2.5) points lower (better) than sham, although also not significant, $p = 0.252$.

Interestingly, in our cohort, we found that measures of rigidity and bradykinesia in the lower limbs, as measured by the respective MDS-UPDRS items, changed significantly between baseline and follow-up in the MicroBurst group while no change was evident in the sham group at six months (changes between baseline and follow-up, Microburst: −2.875 [0.90;4.85], $p = 0.011$, sham: 0.17 [−2.51;2.17] $p = 0.862$). Between-group comparisons of change in end-of-treatment leg bradykinesia and rigidity also favoured the microburst group, although did not achieve statistical significance (−2.32 points [−5.70;1.06], $p = 0.158$).

Patient-reported outcomes of freezing, gait and balance tended to favour the sham group (Table 3). However, only the 39-item Parkinson's Disease Questionnaire (PDQ-39)[20] mobility subscore was significantly different between the two groups, with 29.22 points ([10.21;48.23], $p = 0.008$) estimated difference in favour of the sham group. This difference remained significant after correction for baseline H&Y stage, $p = 0.027$. Furthermore, with adjustment for levodopa equivalent daily dose, comparisons of New Freezing Of Gait Questionnaire (NFOG-Q)[21–23] and timed-up-and-go (TUG) with cognitive dual task (DT) reached significance, favouring sham (NFOG-Q: 4.62 [0.29;8.95] points, $p = 0.04$, and TUG with cognitive DT: 9.63 [1.97;17.30] points, $p = 0.02$). Adjusting for age did not change the results.

Table 1 | Baseline characteristics of the MicroBurst and sham group of the randomised control trial

Characteristics at baseline	Sham (*n* = 6)	MicroBurst (*n* = 6)	*p*-value
Sex, number of females	0	4	
H&Y stage, number of participants stage II/III	6/0	3/3	
Age, years	64.67 ± 5.8	69.33 ± 7.3	0.2483
Disease duration, years	12.33 ± 7.0	8.67 ± 3.3	0.2722
Levodopa equivalent daily dose, mg	1368.50 ± 350	1037.08 ± 356	0.2403
MOCA score, a higher score is better	27.17 ± 1.3	26.0 ± 1.5	0.1918
MDS-UPDRS subscores, lower score is better:			
PIGD score	5.83 ± 2.6	9.33 ± 3.9	0.0977
LBG score	10.8 ± 2.5	17.0 ± 4.6	0.0168 *
Leg bradykinesia and rigidity items	8 ± 2.1	12.16 ± 3.1	0.0205 *
Total motor score	32 ± 9.7	41.8 ± 9.5	0.152
Patient reported gait and balance:			
NFOG-Q, *lower score is better*	21.4 ± 3.8	22.17 ± 5.2	0.7902
ABC, *higher score is better*	63.11 ± 12.8	48.23 ± 17.3	0.1208
PDQ-39 mobility subscore, *lower score is better*	37.92 ± 15.9	52.5 ± 15.1 #	0.1562
Objective gait and balance:			
Berg's Balance Scale, *higher score is better*	52 ± 3.3	48.67 ± 2.94	0.0940
TUG, sec	9.29 ± 1.8	14.19 ± 5.8	0.0942
TUG with cognitive dual task, sec	10.06 ± 1.4	19.99 ± 9.3	0.0478 *
TUG with manual dual task, sec	15.32 ± 8.2	15.93 ± 5.7	0.8850
Modified Figure of Eight Test, sec	71.84 ± 70.2	85.50 ± 50.5	0.7069
20 m walk with regular speed, sec	17.63 ± 3.8	18.75 ± 2.3	0.5545
20 m walk with fast speed, sec	14.24 ± 2.8	14.85 ± 3.5	0.7446
Patient reported quality of life:			
PDQ-39, *lower score is better*	30.74 ± 7.3	24.32 ± 4.3 #	0.1190
SF-36 Physical score, *higher score is better*	43.63 ± 12.3	36.5 ± 11.0 #	0.3426
SF-36 Mental score, *higher score is better*	65.13 ± 12.0	66.74 ± 10.1 #	0.8166

Results are presented as estimated means ±standard deviation, except for sex and Hoehn and Yahr (H&Y) stage. *P*-values derived from two-sided Student's t on the group difference are given. *: Significant difference $p \leq 0.05$. #: $N = 5$. *ABC*: Activities-specific Balance Confidence scale. *LBG*: Lower Body and Gait subscore from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *m*: metres. *MDS-UPDRS*: Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *MOCA*: Montreal Cognitive Assessment. *NFOG-Q*: New Freezing of Gait Questionnaire. *PIGD*: Postural Instability and Gait Difficulty subscore of the MDS-UPDRS. *PDQ-39*: Parkinson's Disease Questionnaire. *sec*: seconds. *SD*: standard deviation. *SF-36*: Short Form Health Survey. *TUG*: Timed-up-and-go. Source data are provided as a Source Data file.

Table 2 | List and distribution of adverse events

Adverse Events	Early start (*n* = 6)	Delayed start (*n* = 6)	Pilots (*n* = 2)	Consequence
Procedural AEs				
Dural puncture	1	0	0	Reoperation (*n* = 1)
Lead misplacement	2	0	0	Reoperation (*n* = 2)
Peri-operative pain	1	0	0	Inpatient additional day (*n* = 1)
Oedema/swelling over the implantation site	3	2	0	Self-limited (*n* = 5)
Peri-surgical increase in Parkinsonian symptoms	3	6	0	Self-limited (*n* = 9)
AEs during the trial period				
Falls requiring hospital contact	1 (One fall during RCT, one during open phase)	1 (One fall during RCT, one during open phase)	1	Two fractures of the proximal humerus (Early start) One Colles fracture (Pilots)

AEs: Adverse events. RCT: Randomised controlled trial. n: number of individuals.

Regrettably, adherence to the protocol for wearing the home-worn device was suboptimal (Supplementary Table 16). No differences between the groups were found at six months (Supplementary Table 17).

Individually, within the MicroBurst group, one participant showed improvement in PIGD score at six months with a clinically significant decrease of three points. This participant, and three additional participants, showed improvements in the LBG score of three to five points. The remaining participants, who were among the oldest of the cohort (71 and 80 years old) and both at H&Y stage III, were largely unchanged, with one point improvement or worsening in LBG score. In the sham group, two participants showed slight improvements of one

Table 3 | RCT group differences in all assessments after six months

Outcome	MicroBurst		Sham		Baseline adjusted differences in means at 6-months (95% CI)	*p*-value
	Baseline mean ± SD	6-month follow-up mean ± SD	Baseline mean ± SD	6-month follow-up mean ± SD		
MDS-UPDRS subscores, lower score is better:						
PIGD score	9.33 ± 3.9	9.0 ± 3.0	5.83 ± 2.6	6.83 ± 3.3	−0.69 (−3.23; 1.85)	0.556
LBG score	17.0 ± 4.6	14.3 ± 5.8	10.8 ± 2.5	11.2 ± 3.3	−2.93 (−8.3; 2.5)	0.252
Total motor score	41.83 ± 9.5	38.83 ± 10.6	32.0 ± 9.7	32.83 ± 8.0	−1.11 (−11.07; 8.84)	0.806
Patient reported gait and balance:						
NFOG-Q score, *lower score is better*	22.17 ± 5.2	24.33 ± 3.4	21.4 ± 3.8	20.5 ± 3.9	3.48 (0.64; 7.59)	0.087
ABC scale, *higher score is better*	48.23 ± 17.3	39.18 ± 17.0	63.11 ± 12.8	59.17 ± 15.8	−15.17 (−39.69; 9.35)	0.195
PDQ-39 mobility sub-score, *lower score is better*	52.5 ± 15.1 #	65.42 ± 12.0	37.92 ± 15.9	37.92 ± 9.9	29.22 (10.21; 48.23)	0.008*
P-GIC at 6 months, a *higher score is better*		1.0 ± 0		1.33 ± 0.5	−0.33 (−0.80; 0.14)	0.145
Objective gait and balance:						
Berg's Balance Scale, *higher score is better*	48.67 ± 2.9	48.33 ± 4.3	52.0 ± 3.3	51.83 ± 4.7	−3.48 (−10.65; 3.69)	0.301
TUG, sec	14.19 ± 5.8	14.75 ± 5.5	9.29 ± 1.8	9.82 ± 2.2	1.93 (−3.47; 7.33)	0.439
TUG with cognitive dual task, sec	19.99 ± 9.3	20.14 ± 7.1	10.1 ± 1.4	11.00 ± 1.7	5.61 (−2.53; 13.75)	0.154
TUG with manual dual task, sec	15.93 ± 5.7	16.72 ± 5.4	15.3 ± 8.2	16.67 ± 5.9	−0.33 (−5.26; 4.61)	0.884
Modified Figure of Eight Test, sec	85.50 ± 50.5	151.72 ± 147.3	71.84 ± 70.2	56.60 ± 37.0	77.03 (−21.97; 176.02)	0.112
20 m regular walk, sec	18.75 ± 2.3	21.71 ± 9.0	17.63 ± 3.8	17.42 ± 3.1	2.84 (−4.65; 10.32)	0.413
20 m fast walk, sec	14.85 ± 3.5	21.14 ± 15.9	14.2 ± 2.8	14.70 ± 2.3	4.98 (−6.83; 16.78)	0.365
Patient reported quality of life:						
PDQ-39, *lower score is better*	24.32 ± 4.3 #	32.09 ± 8.9	30.74 ± 7.3	32.69 ± 14.9	8.44 (−8.39; 25.27)	0.281
SF-36 Physical score, *higher score is better*	36.5 ± 11.0 #	34.92 ± 12.5	43.63 ± 12.3	42.85 ± 19.9 ##	−8.41 (−29.13; 12.30)	0.359
SF-36 Mental score, *higher score is better*	66.74 ± 10.1 #	63.78 ± 18.8	65.13 ± 12.0	53.71 ± 15.3	3.83 (−15.22; 22.87)	0.655
MoCA, *higher score is better*	26.0 ± 1.5	26.83 ± 1.5	27.17 ± 1.3	27.17 ± 2.6	0.39 (−2.54; 3.31)	0.773

Results are presented as the baseline-adjusted between-group difference in end-of-treatment outcome values with a 95% confidence interval. The sham group was set as a reference. *P*-values from the between-group difference in means are presented. *: Significant, α = 0.05. Data obtained were not available for all patients (#: *N* = 5, ##: *N* = 4 indicate the number of participants with available data). *ABC*: Activities-specific Balance Confidence scale. *LBG*: Lower Body and Gait subscore from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *m*: metres. *MDS-UPDRS*: Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *MOCA*: Montreal Cognitive Assessment. *NFOG-Q*: New Freezing of Gait Questionnaire. *P-GIC*: Patient Global Impression of Change. *PIGD*: Postural Instability and Gait Difficulty subscore of the MDS-UPDRS. *PDQ-39*: Parkinson's Disease Questionnaire. *sec*: seconds. *SD*: standard deviation. *SF-36*: Short Form Health Survey. *TUG*: Timed-up-and-go. Source data are provided as a Source Data file.

point in the PIGD score. One of these participants showed the only improvement in LBG score among sham participants, with a six-point decrease. Individual RCT data are detailed in Supplementary Tables 1, 2.

At the end of the open extension phase, compared with pre-stimulation, delayed starters ($n = 5$) showed a significant decrease (improvement) in group mean LBG score by −2.8 points ([−4.64;−0.96], $p = 0.014$) and non-significant, small decreases in PIGD score: −1.8 points [−4.19;0.59], lower body rigidity and bradykinesia items: −1.4 points [−4.39;1.59], and MDS-UPDRS total motor score: −4.2 [−11.51;3.11] at 12 months follow-up.

Following twelve months of SCS stimulation, there was a significant improvement in LBG score (−5.25 points [−8.93; −1.57], $p = 0.012$) and lower body bradykinesia and rigidity items (−3.88 points [−7.17;−0.58], $p = 0.027$, in the early start group ($n = 8$)). Unfortunately, the MDS-UPDRS Part II for one of these patients was not available, and we were not able to determine the PIGD score in this patient. The mean PIGD score of the remaining seven participants did not improve significantly compared with baseline (−0.29 points [−2.72;2.14], $p = 0.783$). Apart from an improvement in Montreal Cognitive Assessment (MoCA)[24] score (26.0 ±1.4 to 27.38 ±1.8, $p = 0.036$), no other assessments showed statistically significant changes after twelve months' stimulation (Table 4, Supplementary Table 18).

We then used a gait-matched group of ten patients from the Parkinson Progression Marker Initiative (PPMI) database[16] (mean age 66.77 ± 9.9, 1 female/9 males) as an external comparator arm. The patients matched our cohort on PIGD (PPMI: 6.40 ± 3.7 vs STEP-PD: 7.62 ± 3.5, $p = 0.431$) and LBG (PPMI: 12.10 ± 4.1 vs STEP-PD: 14.15 ± 4.7, $p = 0.284$) scores at baseline. Interestingly, while the PPMI patients showed disease progression with worsening of PIGD score over one year, the patients with active SCS in our cohort tended to improve over the same period of time (Fig. 2). The estimated, pre-stimulation-adjusted PIGD score of STEP-PD participants was −1.66 points (−3.50;0.19) lower compared with PPMI participants; however, not

Table 4 | Within-group comparisons, twelve months of active SCS-treatment

Outcome	Baseline (*n* = 8) mean ± SD	12-month follow-up (*n* = 8) mean ± SD	Difference (95%CI)	*p*-value, paired *t*-test
H&Y stage	2.63 ± 0.5	2.38 ± 0.5	−0.25 (−0.64; 0.14)	0.1705
MOCA score, *higher score is better*	26.0 ± 1.4	27.38 ± 1.8	1.375 (0.12; 2.63)	0.0363 *
MDS-UPDRS subscores, lower score is better:				
PIGD score (*n* = 7)	8.29 ± 2.6	8.0 ± 3.5	−0.29 (−2.72; 2.14)	0.7833
LBG score	16.38 ± 4.3	11.13 ± 4.4	−5.25 (−8.93; −1.57)	0.0118 *
Leg bradykinesia and rigidity items	11.63 ± 3.2	7.75 ± 3.2	−3.88 (−7.17; −0.58)	0.0273 *
Total motor score	40.50 ±10.1	34.88 ± 11.0	−5.63 (−14.03; 2.78)	0.1576
Patient reported gait and balance:				
NFOG-Q score, (*n* = 6) *lower score is better*	20.50 ± 4.9	19.5 ± 6.3	−1.00 (−5.30; 3.30)	0.5761
ABC scale, (*n* = 7) *higher score is better*	58.45 ± 12.5	53.72 ± 18.0	−4.73 (−13.85; 4.38)	0.2510
PDQ-39 mobility subscore, (*n* = 7) *lower score is better*	53.21 ± 12.7	58.57 ± 10.6	5.36 (−6.39; 17.10)	0.3071
Objective gait and balance:				
Berg's Balance Scale, *higher score is better*	48.63 ± 4.3	51.13 ± 4.2	2.50 (−1.50; 6.50)	0.1827
TUG, sec.	12.68 ± 5.8	13.53 ± 5.3	0.85 (−3.41; 5.10)	0.6513
TUG, with cognitive dual task sec.	17.38 ± 9.4	16.13 ± 8.1	−1.25 (−6.64; 4.14)	0.5993
TUG, with manual dual task, sec.	14.32 ± 5.8	13.90 ± 5.7	−0.42 (−3.04; 2.20)	0.7134
Modified Figure of Eight Test, sec., (*n* = 7)	80.46 ± 48.0	128.37 ± 103.6	47.91 (−29.83; 125.66)	0.1823
20 m regular walk, sec.	18.93 ± 2.3	19.25 ± 5.5	0.32 (−3.01; 3.65)	0.8269
20 m fast walk, sec.	14.74 ± 3.9	16.65 ± 5.1	1.91 (−1.21; 5.03)	0.1919
Patient reported quality of life:				
PDQ-39, (*n* = 7) *lower score is better*	28.90 ± 9.1	31.34 ± 14.5	3.44 (−4.12; 11.00)	0.3083
SF-36 Physical score, (*n* = 6) *higher score is better*	37.64 ± 9.6	26.42 ± 9.6	−11.22 (−26.25; 3.82)	0.1133
SF-36 Mental score, (*n* = 6) *higher score is better*	59.89 ± 14.2	59.38 ± 19.3	−0.51 (−24.68; 23.65)	0.9587

Results are presented as estimated means ±standard deviation and mean difference with p-values derived from a two-sided, paired *t*-test. Eight individuals were included in the analyses unless otherwise specified. *: Significant difference $p \leq 0.05$. *ABC*: Activities-specific Balance Confidence scale. *LBG*: Lower Body and Gait subscore from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *m*: metres. *MDS-UPDRS*: Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *MOCA*: Montreal Cognitive Assessment. *NFOG-Q*: New Freezing of Gait Questionnaire. *PIGD*: Postural Instability and Gait Difficulty subscore of the MDS-UPDRS. *PDQ-39*: Parkinson's Disease Questionnaire. *sec*: seconds. *SD*: standard deviation. *SF-36*: Short Form Health Survey. *TUG*: Timed-up-and-go. Source data are provided as a Source Data file.

statistically significant, $p = 0.075$. Similar trends were observed for the mean LBG score and MDS-UPDRS total motor score (Supplementary Table 19).

As a secondary analysis, comparison of the changes from the pre-stimulation assessment to end-of-trial follow-up in all our participants ($n = 13$: one year's change in eight early starters and six months' change in five delayed starters) was performed. This analysis showed no significant improvement in PIGD score (−0.92 points [−2.44;0.60], $p = 0.211$). However, for the reason explained above, the PIGD score was only available for 12 patients. Conversely, the LBG score in all 13 patients improved by −4.31 points (−6.35;−2.08), $p = 0.001$, which was driven by improvement in bradykinesia and rigidity items (−2.92 [−5.07;−0.78], $p = 0.012$). No other clinical outcomes differed significantly (Table 5, Supplementary Table 20).

Finally, we performed an exploratory analysis on all the participants who had received six months of SCS stimulation ($n = 13$), pooling data from the first six months of stimulation of the eight early starters and the six months of active stimulation in the five delayed starters. After six months of SCS stimulation, the PIGD score showed a statistically significant decrease (−1.31 points [−2.47;−0.14], $p = 0.031$, $n = 13$). Thirty percent of these patients ($n = 4$) had a clinically meaningful improvement of PIGD score of at least three points (−3.75 points [−5.27;−2.23], $p = 0.004$). During the first six months of stimulation ($n = 13$), the LBG score improved by −3.23 points (−4.52;−1.95), $p = 0.0001$, a decrease that was especially driven by improvement in lower body rigidity and bradykinesia items (−2.31 points [−3.75;−0.86], $p = 0.005$). The MDS-UPDRS total motor score improved by −4.23 points (−7.62;−0.84), $p = 0.019$). However, despite this objective improvement, patients reported a deterioration in the PDQ-39 (5.24 points [0.91;9.58], $p = 0.0221$) driven by items of the mobility, activities of daily living, and cognition subscores. The remaining clinical outcomes did not differ significantly in six months (Table 6, Supplementary Table 21).

At the end-of-trial, eleven participants answered the Patient Global Impression of Change questionnaire, with 45% ($n = 5$) stating that SCS had improved their gait function with a meaningful impact, and a further 36% ($n = 4$) felt some improvement, however, not a certain meaningful impact on gait function (Supplementary Table 20). Four out of the five participants who reported an impactful improvement were early starters with SCS stimulation for twelve months. Their stimulation paradigms at end-of-trial were a mixture of MicroBurst ($n = 3$) and square-wave ($n = 2$). These participants had group mean

https://doi.org/10.1038/s41467-026-68782-w

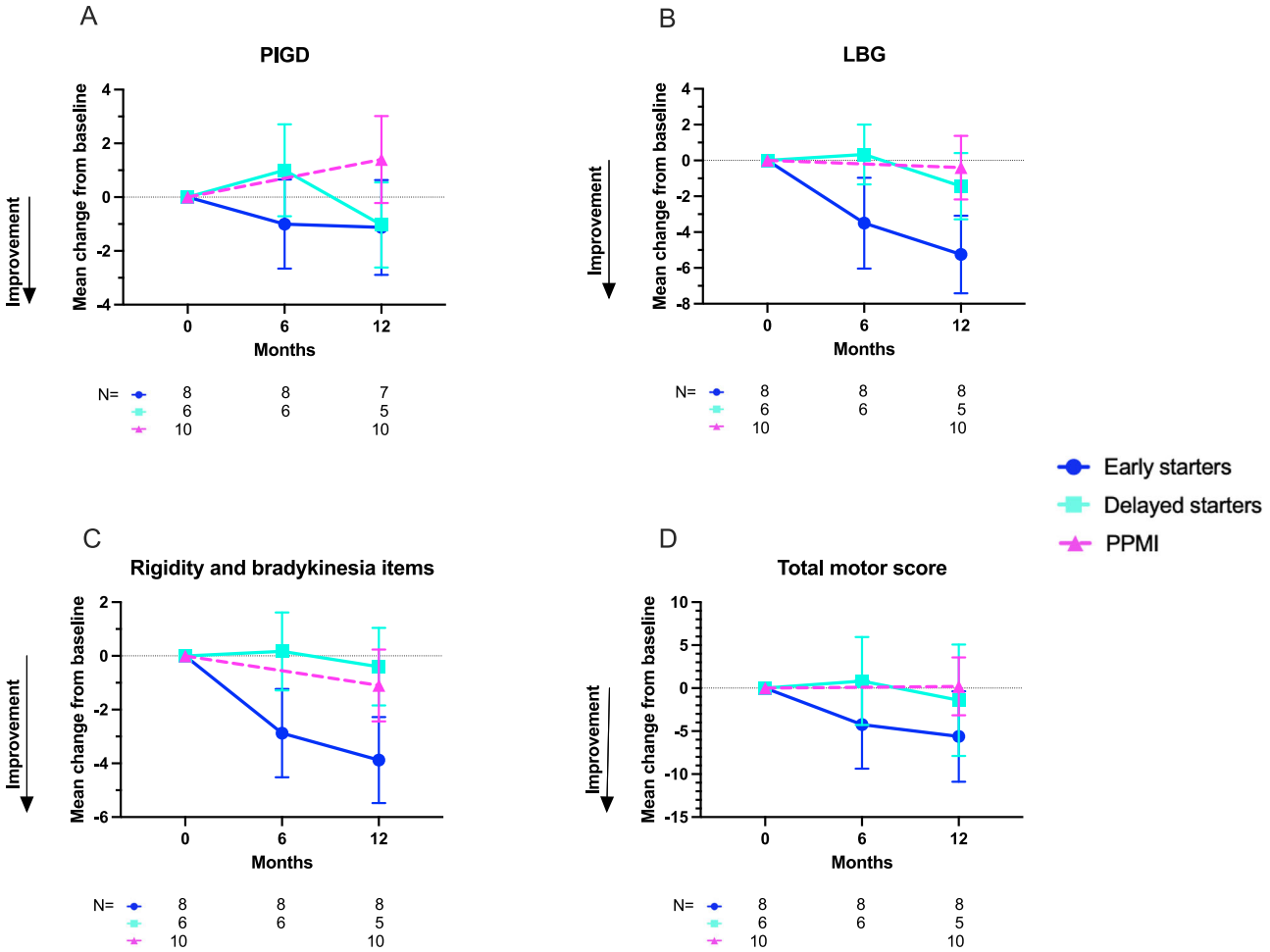

Fig. 2 | Changes in MDS-UPDRS subscores. Mean changes from baseline with standard errors of Movement Disorder Society-Unified Parkinson's Disease Rating Scale (MDS-UPDRS) subscores; **A** the Postural Instability and Gait Disability (PIGD) score, **B** the Lower Body and Gait (LBG) score, **C** summed rigidity and bradykinesia items, and **D** total motor score. Decreases marks improvement. Groups are divided into early starters (dark blue) with twelve months of SCS, delayed starters (light blue) with active stimulation only after six months, and participants from the Parkinson's Progression Markers Initiative (PPMI) with data at years seven and eight (pink) who were matched to our cohort on gait impairment and the presence of freezing of gait. Number of included individuals at each time point is presented below graphs. Source data are provided as a Source Data file.

improvements in PIGD, LBG, and rigidity and bradykinesia items compared with pre-stimulation baseline; however, only the difference in LBG score reached statistical significance (−4.8 points [−9.56;−0.04], $p = 0.049$).

At the time of writing, eight participants have so far opted for a second surgical procedure to change their IPG when its battery was depleted (all battery replacements were necessitated only after the conclusion of the trial period). Four participants have had their device removed; one person withdrew during the open extension of the study, and one opted for removal of the system after 1.5 years of treatment—both due to lack of subjective improvement. Two participants were deemed unfit for IPG change due to an overall deterioration in function and co-morbidity, one of whom had had a previous second IPG.

Recently, three of the patients were considering whether to choose an IPG change (first or second) and reported experiencing a notable worsening in FoG after three to four weeks of the stimulation being turned off. Consequently, they opted to continue SCS.

Imaging assessments

As to the imaging component of the trial, [^{18}F]-FDG and [^{18}F]-FEOBV PET imaging was successfully performed at baseline and two follow-ups for thirteen participants, while one only had the first two scans due to withdrawal as described above. [^{18}F]-FDG mean injection doses of 187 MBq ±11.6, 185.6 MBq ±8.3 and 185.8 ± 5.9 MBq were administered at baseline, 6-month and 12-month visits, respectively. [^{18}F]-FEOBV mean injection doses of 200 MBq ±12.6, 206.7 MBq ±15.6 and 200.8 ± 7.4 MBq were administered at baseline, 6-month and 12-month visits, respectively. In the interest of imaging processing homogeneity, one participant (pilot) was subsequently excluded from imaging analyses due to a lack of an MRI for spatial segmentation. Nine controls with idiopathic PD and no gait impairment (mean age 66.78 years ±7.2, mean disease duration 7.7 years ±2.2, all H&Y stage II, 4 females) had comparable [^{18}F]-FEOBV PET images with a mean injection dose of 207.5 MBq ±19.8.

In regard to the double-blinded phase, SPM analyses comparing the MicroBurst ($n = 6$) and sham ($n = 6$) groups after the RCT showed no significant differences in glucose metabolism or in [^{18}F]-FEOBV uptake when adjusted for baseline scans. Adjusting for age, disease severity and levodopa equivalent daily dose did not yield any changes.

When evaluating the effect of active stimulation overall, at baseline, our gait-impaired participants had a higher [^{18}F]-FEOBV uptake in the right anterior insula and frontal cortex when compared with non-gait-impaired PD patients ($p = 0.001$, $\alpha < 0.001$, Fig. 3, Supplementary Table 22).

Comparison of pre-active stimulation baseline with the final trial follow-up of our entire cohort (one-year stimulation $n = 7$, six-month

Table 5 | Within-group comparisons, all participants pre-stimulation assessment to 12-month follow-up

Outcome	Baseline (*n* = 13) mean ± SD	12-month follow-up (*n* = 13) mean ± SD	Difference (95%CI)	*p*-value, paired *t*-test
H&Y stage	2.38 ± 0.5	2.23 ± 0.4	−0.15 (−0.38; 0.07)	0.1654
MOCA score, *higher score is better*	26.77 ± 1.8	27.38 ± 1.9	0.62 (−0.53; 1.76)	0.2643
MDS-UPDRS subscores, lower score is better:				
PIGD score (*n* = 12)	7.58 ± 3.0	6.67 ± 3.4	−0.92 (−2.44; 0.60)	0.2112
LBG score	14.77 ± 4.1	10.46 ± 4.0	−4.31 (−6.53; −2.08)	0.0012 *
Leg bradykinesia and rigidity items	10.62 ± 3.1	7.69 ± 2.9	−2.92 (−5.07; − 0.78)	0.0117 *
Total motor score	38.31 ± 9.2	33.23 ± 11.0	−5.08 (−10.17; 0.02)	0.0507
Patient reported gait and balance:				
NFOG-Q score, (*n* = 11) *lower score is better*	20.00 ± 4.1	17.82 ± 5.9	−2.18 (−5.42; 1.05)	0.1636
ABC scale, (*n* = 10) *higher score is better*	59.79 ± 12.5	54.48 ± 16.9	−5.31 (−11.19; 0.57)	0.0713
PDQ-39 mobility subscore, (*n* = 12) *lower score is better*	45.83 ± 14.2	48.33 ± 16.0	2.5 (−3.89; 8.89)	0.4075
Objective gait and balance:				
Berg's Balance Scale, *higher score is better*	50.54 ± 4.3	52.00 ± 4.0	1.46 (−1.16; 4.08)	0.2472
TUG, sec.	11.58 ± 4.9	12.43 ± 4.7	0.85 (−1.58; 3.27)	0.4608
TUG, with cognitive dual task sec.	14.92 ± 7.9	14.56 ± 7.1	−0.36 (−3.69; 2.97)	0.8185
TUG, with manual dual task, sec.	14.99 ± 5.8	13.65 ± 5.1	−1.34 (−3.78; 1.10)	0.2546
Modified Figure of Eight Test, sec., (*n* = 12)	71.72 ± 44.4	92.86 ± 88.4	21.14 (−26.42; 68.69)	0.3490
20 m regular walk, sec.	18.14 ± 2.7	18.13 ± 4.7	−0.003 (−1.94; 1.93)	0.9973
20 m fast walk, sec.	14.45 ± 3.1	15.46 ± 4.4	1.01 (−1.03; 3.05)	0.3011
Patient reported quality of life:				
PDQ-39, (*n* = 12) *lower score is better*	29.04 ± 10.7	31.91 ± 13.2	2.87 (−0.92; 6.67)	0.1237
SF-36 Physical score, (*n* = 9) *higher score is better*	40.27 ± 14.5	32.80 ± 14.3	−7.47 (−17.85; 2.91)	0.1356
SF-36 Mental score, (*n* = 11) *higher score is better*	56.33 ± 15.0	55.85 ± 18.1	−0.47 (−11.67; 10.73)	0.9269

Results are presented as estimated means ±standard deviation and mean difference with p-values derived from a two-sided, paired *t*-test. Thirteen individuals were included in the analyses unless otherwise specified. *: Significant difference $p \leq 0.05$. *ABC*: Activities-specific Balance Confidence scale. *LBG*: Lower Body and Gait subscore from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *m*: metres. *MDS-UPDRS*: Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *MOCA*: Montreal Cognitive Assessment. *NFOG-Q*: New Freezing of Gait Questionnaire. *PIGD*: Postural Instability and Gait Difficulty subscore of the MDS-UPDRS. *PDQ-39*: Parkinson's Disease Questionnaire. *sec*: seconds. *SD*: standard deviation. *SF-36*: Short Form Health Survey. *TUG*: Timed-up-and-go. Source data are provided as a Source Data file.

stimulation $n = 5$), showed significant bilateral decrease in thalamic glucose metabolism ($\alpha < 0.01$, $p = 0.005$, Fig. 4, Supplementary Table 23) as well as a decrease in [^{18}F]-FEOBV uptake in the right anterior insula ($\alpha < 0.001$, $p < 0.001$).

Analysis of one year's treatment (n = 7) also showed a significant decrease in glucose metabolism in one cluster that included the left amygdala and parahippocampus ($\alpha < 0.01$, $p = 0.029$), and the thalamus ($\alpha < 0.05$, $p < 0.001$). After one year's treatment, the [^{18}F]-FEOBV uptake showed significant decreases in right motor cortex ($\alpha < 0.01$, $p < 0.001$), right supplementary motor cortex ($\alpha < 0.01$, $p = 0.015$), left ventral diencephalon ($\alpha < 0.01$, $p < 0.001$), and right anterior insula ($\alpha < 0.01$, $p < 0.001$, Fig. 3, Supplementary Table 24).

An exploratory analysis of the [^{18}F]-FDG PET images comparing the first six months of stimulation in all participants (n = 12) showed a large cluster that included the thalamus bilaterally ($\alpha < 0.05$, $p = 0.052$). However, no significant changes in [^{18}F]-FEOBV uptake were detected in this group.

Discussion

This study demonstrates that conducting an RCT of MicroBurst SCS in gait-impaired PD patients is feasible, and no major safety issues were reported. We achieved a recruitment rate of 1.2 participants per month at a single site. All participants completed the RCT and adhered to their allocated group. Only 93 % completing the entire one-year trial, and 6/13 participants changed their paradigm from MicroBurst to square-wave stimulation three months before end-of-trial. Improving the recruitment rate may be achieved in future studies by broadening recruitment sources in multi-centre trials. Furthermore, while our enrolment criteria secured well-characterised participants who were also eligible for PET imaging, a higher recruitment rate may also be achieved in trials that only aim to assess SCS efficacy.

STEP-PD confirmed SCS to be a safe and acceptable therapy for patients with PD and gait impairment with no stimulation-induced adverse events. The most reported AE was a transient increase in Parkinsonian symptoms following surgery, which did not require medication changes. This worsening could be related to the effect of surgery and the mobility restrictions in the six-week post-operative period. The implementation of guided rehabilitation or physical therapy in the post-operative period could minimise this side effect and help mitigate the psychological effect of this temporary worsening of Parkinsonian symptoms in participants of future trials. This approach has successfully been used in other trials of SCS for other indications in non-PD patients[25].

Table 6 | Within-group comparisons, six months of active SCS-treatment

Outcome	Baseline (*n* = 13) mean ± SD	6-month follow-up (*n* = 13) mean ± SD	Difference (95%CI)	*p*-value, paired *t*-test
H&Y stage	2.38 ± 0.5	2.38 ± 0.7	0 (−0.35; 0.35)	1.0000
MOCA score, *higher score is better*	26.77 ± 1.8	27.08 ± 1.6	0.31 (−0.66; 1.28)	0.5016
MDS-UPDRS subscores, lower score is better:				
PIGD score	8.15 ± 3.6	6.85 ± 3.4	−1.31 (−2.47; −0.14)	0.0311 *
LBG score	14.77 ± 4.1	11.54 ± 5.2	−3.23 (−4.52;−1.95)	0.0001 *
Leg bradykinesia and rigidity items	10.62 ± 3.1	8.31 ± 3.1	−2.31 (−3.75;− 0.86)	0.0046 *
Total motor score	38.31 ± 9.2	34.08 ± 10.9	−4.23 (−7.62; −0.84)	0.0187 *
Patient reported gait and balance:				
NFOG-Q score, (*n* = 13) *lower score is better*	20.00 ± 4.7	20.31 ± 5.6	0.31 (−2.65; 3.26)	0.8243
ABC scale, (*n* = 11) *higher score is better*	56.12 ± 17.0	47.45 ± 17.2	−8.67 (−21.29; 3.96)	0.1571
PDQ-39 mobility subscore, (*n* = 12) *lower score is better*	45.83 ± 14.2	51.67 ± 18.7	5.83 (-−3.26; 14.93)	0.1857
Objective gait and balance:				
Berg's Balance Scale, *higher score is better*	50.54 ± 4.3	50.69 ± 4.2	0.15 (−2.61; 2.92)	0.9056
TUG, sec.	11.58 ± 4.9	12.13 ± 4.9	0.55 (−1.19; 2.30)	0.5016
TUG, with cognitive dual task sec.	14.92 ± 7.9	15.45 ± 7.2	0.52 (−3.13; 4.18)	0.7597
TUG, with manual dual task, sec.	14.99 ± 5.8	14.34 ± 5.3	−0.65 (−3.08; 1.78)	0.5710
Modified Figure of Eight Test, sec., (*n* = 12)	71.72 ± 44.4	97.94 ± 114.2	26.21 (−26.43; 78.85)	0.2965
20 m regular walk, sec.	18.14 ± 2.7	19.14 ± 6.6	1.00 (−2.21; 4.22)	0.5106
20 m fast walk, sec.	14.45 ± 3.1	17.24 ± 11.1	2.78 (−2.88; 8.44)	0.3050
Patient reported quality of life:				
PDQ-39, (*n* = 12) *lower score is better*	29.04 ± 10.7	34.28 ± 11.8	5.24 (0.91; 9.58)	0.0221 *
SF-36 Physical score, (*n* = 10) *higher score is better*	38.84 ± 14.4	35.50 ± 11.1	−3.35 (−10.58; 3.89)	0.3227
SF-36 Mental score, (*n* = 12) *higher score is better*	57.62 ± 15.0	49.88 ± 19.4	−7.74 (−17.00; 1.52)	0.0930

Results are presented as estimated means ±standard deviation and mean difference with p-values derived from a two-sided, paired *t*-test. Thirteen individuals were included in the analyses unless otherwise specified. *: Significant difference $p \leq 0.05$. *ABC*: Activities-specific Balance Confidence scale. *LBG*: Lower Body and Gait subscore from the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *m*: metres. *MDS-UPDRS*: Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *MOCA*: Montreal Cognitive Assessment. *NFOG-Q*: New Freezing of Gait Questionnaire. *PIGD*: Postural Instability and Gait Difficulty subscore of the MDS-UPDRS. *PDQ-39*: Parkinson's Disease Questionnaire. *sec*: seconds. *SD*: standard deviation. *SF-36*: Short Form Health Survey. *TUG*: Timed-up-and-go. Source data are provided as a Source Data file.

No significant impact on gait impairment was observed during the double-blind phase. Neither the PIGD score nor the LBG score improved significantly with SCS compared to sham. There are several factors that could have affected the results of the double-blind phase and should be taken into account when designing future RCT of SCS in PD. Firstly, the small sample size of this study. While this was an initial feasibility study and the clinical outcome was not the main aim of the study, we had estimated that a sample size of 12 participants would provide us with 96% power to detect a three-point difference in mean changes from baseline in PIGD score ($p < 0.05$) between groups. This power calculation was done using the SD of normal progression over one year in 709 patients with PD with gait problems in the PPMI database. However, the PPMI cohort consists of a more homogeneous population in terms of disease duration and severity of gait problems compared to our participants. The high variability in age, disease duration, H&Y stage, and severity of gait problems in our sample could have reduced our statistical power to detect statistically significant differences between groups. Hopefully, the SDs reported in this paper could help more precise calculation of sample sizes needed in future trials. Secondly, patients were consecutively randomised using a 1:1 ratio but without a perfect match of the patients in the two groups. While this was done to optimise enrolment and the duration of the trial, it resulted in a slight mismatch of the two groups, as the MicroBurst group was, in general, slightly worse overall (Table 1) and included all the patients with H&Y stage III, the oldest patient in the study, and all the patients who had suffered surgical complications requiring reoperation. At the individual level, within the MicroBurst RCT group, the oldest patients with H&Y stage III had generally worse outcomes. Furthermore, the distribution of males and females was unbalanced between groups. Using a cross-over design could have mitigated the population mismatch. However, the impact of surgery in both arms during the first phase of a cross-over trial could be a confounding factor. Therefore, in future studies, a more careful stratification of the population during randomisation and longer double-blind phases should be considered, particularly if small sample sizes are used. Finally, the patients' expectations, which were often very high, were not met, and this could have also played a role. This could also explain the discrepancy between objective and subjective outcomes that we observed at six months.

An intriguing result of this trial was the positive effect of SCS on bradykinesia and rigidity in the legs after both six months and twelve months of stimulation. SCS has previously been reported to be able to alleviate spasticity across a range of dyskinetic movement disorders of central origin[26–28]. Our findings of significant reductions within-group of rigidity and bradykinesia of the lower limbs after SCS are, therefore, promising given the positive correlation previously described between severity of rigidity and FoG[29]. It would also suggest that the LBG score, which includes MDS-UPDRS subscores for rigidity and bradykinesia of the lower limbs, could be a more appropriate outcome measure than PIGD to assess the effect of SCS on gait problems in PD in future trials and could also serve as an additional selection parameter in these studies. In fact, significant changes in LBG score after stimulation were observed at twelve months, and in the comparisons between pre-stimulation to end-of-trial assessment and pre-stimulation to six-

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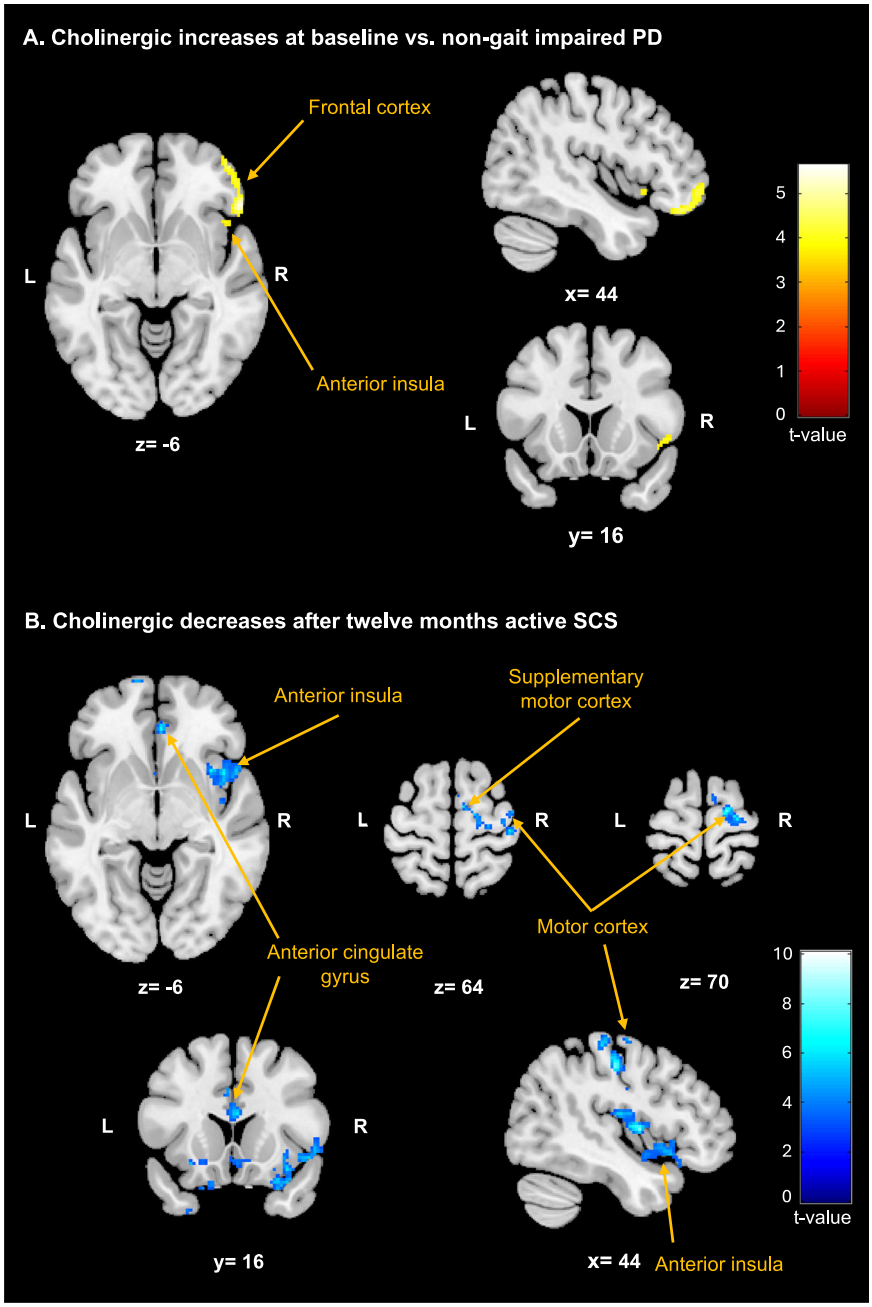

Fig. 3 | Cholinergic changes. A Two-group voxel-wise analysis of [^{18}F]-FEOBV PET in our participants compared with non-gait impaired PD patients showed increased uptake in a cluster including the frontal cortex and the right anterior insula ($\alpha < 0.001$, $p < 0.5$). **B** Paired voxel-wise analysis of [^{18}F]-FEOBV PET of twelve months SCS compared with baseline in seven participants showed decreases in several regions, including the right anterior cingulate gyrus, anterior insula, motor cortex, and supplementary motor area ($\alpha < 0.01$, $p < 0.5$). x,y,z: coordinates in the MNI space on the left-right, anterior-posterior, and inferior-superior axes, respectively. All statistical tests were two-sided, and *p*-values are family-wise error-corrected.

month stimulation in the whole cohort, whereas significant changes in PIGD score were only observed in the latter comparison with a larger number of patients (−1.31 [−2.47;−0.14], $p = 0.0311$), possibly suggesting that PIGD score is less sensitive to changes in small sample sizes.

With respect to the LBG score and leg rigidity and bradykinesia scores, it should also be noted that the observed improvement in these measures increased with stimulation time. This suggests that SCS may have some delayed effects on these symptoms and motor circuits regulating gait.

Remarkably, when we paralleled our participants to a gait-matched group from the PPMI database as an external comparator arm, we observed a worsening in PIGD and LBG scores in the PPMI patients over one year, whereas the patients with early start active SCS tended to improve over the same period. The delayed starters had an initial deterioration of scores during the six months of sham stimulation, followed by changes in the opposite direction when switched to active stimulation (Fig. 2). While the results of this exploratory analysis did not reach statistical significance, they appear to provide further support to the concept that long-term SCS treatment might be needed to improve gait symptoms in PD.

At the end-of-trial, five (45%) of the eleven participants who answered the Patient Global Impression of Change questionnaire reported that SCS had improved their gait function with a meaningful impact. Four of them were early starters with SCS stimulation for

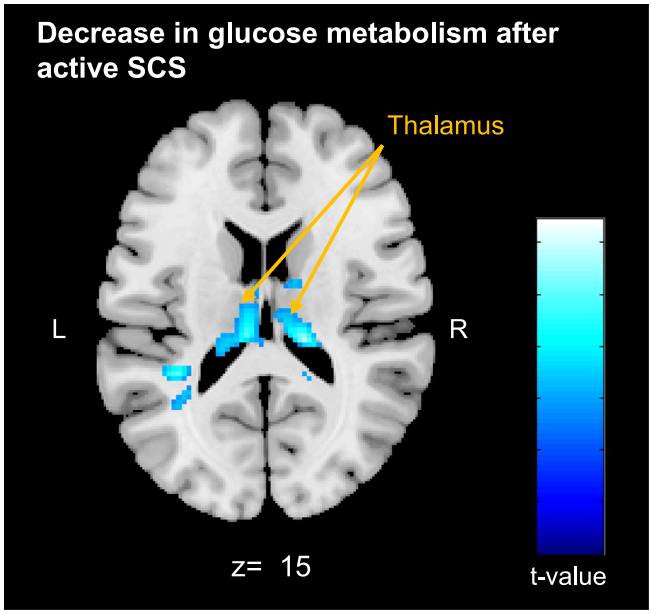

Fig. 4 | Metabolic changes. Paired voxel-wise comparison of pre-active stimulation [^{18}F]-FDG PET imaging with final follow-up in twelve participants showed a bilateral decrease in thalamic metabolism (Family-wise error-corrected $p = 0.005$). z: coordinate in the MNI space on the inferior-superior axis. Two-sided statistical test ($\alpha < 0.01$).

twelve months, providing further support to the need for long-term treatment. Apart from that, we were not able to identify in these patients any trait/factor that could predict a more favourable response to SCS. Larger cohorts of patients are required to further investigate whether some PD patients are more prone to respond to SCS than others.

Patient-reported outcomes improved in the unblinded, open phase, and a placebo effect can therefore not be excluded. However, in corresponding objective outcome measures (Fig. 2), early starters showed their greatest improvements during the double-blinded RCT phase with sustained benefits during the open phase, suggesting that the patient-reported outcomes might not be purely placebo-driven.

In our participants, active SCS was associated with decreased glucose metabolism in the thalamic nuclei, which is consistent with previously reported changes in PD patients with improved rigidity after one year's treatment with subthalamic deep brain stimulation[30]. Indeed, the abnormal metabolic motor network in PD, a pattern including increased metabolism in the thalamus, globus pallidus/putamen, pons, cerebellum, as well as hypometabolism in the premotor cortex and parieto-occipital associated regions, has been associated with bradykinetic-rigid Parkinsonian symptoms in several cohorts[31,32]. SCS in animal models of PD, including non-human primates, has shown improvement in locomotion by activating the dorsal column-medial lemniscal pathway, which in turn desynchronizes abnormal cortico-striatal oscillations[33,34]. Inputs from ascending lemniscal and extralemniscal pathways to the brainstem and thalamus may modulate the supplementary motor area[35]. Thus, SCS might modulate thalamic activity and thereby leg rigidity and bradykinesia through these pathways.

It could be argued that the observed metabolic decreases at follow-up could be related to a general decrease of [^{18}F]-FDG uptake in PD patients with FoG. Since the significant decrease in[^{18}F]-FDG uptake from baseline to post-SCS in our cohort is not paralleled by a worsening of FoG of gait over the same period, it is unlikely that it is related to the FoG itself. However, this cannot be completely excluded with this study and should be further investigated in future studies.

Finally, at baseline, our participants had increased [^{18}F]-FEOBV uptake in right frontal areas and the attention-regulating right anterior insula compared to a control group of PD patients without gait problems, possibly indicating compensatory mechanisms in PD with gait problems. Active SCS was associated with decreased [^{18}F]-FEOBV uptake—in the anterior insula, the motor cortex, and the supplementary motor area, possibly indicating, at least in the insula, a normalisation of compensatory overexpression of cholinergic nerve function. Interestingly, increased gait-induced activation in the right insula has been reported in PD patients when walking on a treadmill compared to healthy control subjects[36]. This increased activation has been suggested to represent a possible compensatory mechanism in PD, since anterior insula overactivity has also been observed in the post-stroke recovery of motor function[37]. Furthermore, the right anterior insula and supplementary motor area have previously been suggested to be involved in compensatory mechanisms during anticipatory postural adjustments in PD patients with FoG[38].

It should be noted that the follow-up PET scans were co-registered to the baseline MRI, as the SCS device was not compatible with 3 T MRI at the time. However, during one-year, we do not expect any changes at a level that could impair co-registration, nor did our participants suffer any such acute events. PET imaging was included in our study design as an exploratory outcome, and future clinical multi-centre studies should not include PET imaging as a primary outcome.

There are several additional considerations that deserve further discussion, as they could help design future studies.

First, we observed heterogeneous compliance with using the wearable device. Additionally, as participants were instructed not to wear the device during strenuous activity such as training sessions, differences in overall activity levels cannot be discerned. In this trial, the wearable device was included to detect and explore changes in specific features of gait, such as FoG, rather than measuring overall physical activity. In future trials, it may be prudent to also measure overall activity throughout the day, thus not requiring participants to remember to wear the device. Furthermore, future research studies should aim to incorporate objective methods such as kinematic, electrophysiological or electromyographic measures to detect possible improvements in gait and provide useful knowledge on the SCS mechanism of action.

Secondly, our RCT results pertain exclusively to MicroBurst-SCS and may not reflect the efficacy of other stimulation paradigms previously used for SCS[10,11,39–41]. Indeed, the improvements reported from our open-phase contain a mixture of participants with MicroBurst- and square-wave-SCS. Therefore, other stimulation paradigms should be further investigated with appropriate RCTs in new patients but also in PD patients who have already received SCS for gait problems (e.g., cross-over studies with different types of stimulation, as we have started in some of the patients in our cohort who have completed the extension phase). During the open phase, we had to change the stimulation paradigm in some participants, which suggests that MicroBurst might not be the optimal stimulation for all patients and furthermore, that one paradigm may not be right for everyone. Indeed, we are currently conducting a subsequent trial on this cohort, comparing different paradigms.

Thirdly, the majority of our participants experienced self-limiting, post-surgery worsening in general PD symptoms, and future clinical trials might exclude any surgery-related effects by adding a second baseline assessment post-surgery.

Finally, in the STEP-PD trial, final placement was namely between T7-T11, as this was the area mainly used in most of the previous open-label studies, and to preserve standardisation. However, a recent case report has shown that advanced personalised spinal electrical stimulation with an electrode array placed over the lumbar L2 and L4-5 achieved substantial improvement of gait dysfunction in a PD patient[32]. Conceptually, the study was designed to deliver stimulation actively

mimicking spinal patter generation to activate specific motor pools in a time-locked manner. While this study differs from ours in several important respects in terms of rationale, programming, stimulation paradigm, and hardware, it does raise an important question regarding alternative placements of leads for SCS, including stimulation of descending pathways in addition to ascending as done here. Gait is, indeed, a dynamic process in terms of physiology and biomechanics, involving both spinal circuitries and supraspinal control.

In conclusion, the STEP-PD trial provides evidence that SCS applied over thoracic spinal levels is feasible, safe, and well-tolerated by patients with PD, and may, especially over time, provide improvement of bradykinesia and rigidity in the legs and LBG score. However, since the primary clinical outcome, the PIGD score, did not improve significantly during the double-blind phase, this trial did not provide evidence of a clinically relevant improvement in gait and FoG by MicroBurst SCS in PD. Deconditioning seen in PD with progressive gait symptoms is hypothesised to be a significant covariate to be considered in the design of future trials. Furthermore, questions remain regarding placement and stimulation settings, which should be fully explored in larger, well-defined cohorts of PD patients with gait problems.

Method

STEP-PD consisted of a six-month open-label pilot study in two initial pilot patients and a six-month prospective, double-blinded RCT in 12 patients, followed by a six-month open extension phase in all 14 patients. The study protocol has been previously published[14]. All parts of the trial protocol were reviewed and approved by the Scientific Ethics Committee for the Central Denmark Region (Approval 1-10-72-44-21) and registered with clinicaltrials.gov (NCT05110053) prior to data collection (https://clinicaltrials.gov/study/NCT05110053#more-information) on September 23. 2021. Neither the granting body of the study nor the manufacturers of devices used in this trial had any role in the conception of the study, collection, analysis, interpretation, or presentation of resultant data.

Patient recruitment and selection

Patients were recruited through the Neurology Clinic at Aarhus University Hospital, referrals from other national neurologists, or patients' self-referral following advertisement in local and national news and social media. Aarhus University Hospital is a highly specialised centre for advanced treatment of PD as well as refractive pain conditions with SCS. Patients were pre-screened by telephone and, if suitable, fully assessed in subsequent visits to the clinic. Recruitment started in November 2021, and the last patient was recruited on October 1st, 2022. All participants gave written, informed consent.

Participants were evaluated for the following inclusion criteria: Idiopathic PD diagnosed by a movement disorders neurologist according to the MDS criteria[17], presence of FoG according to the first question of the NFOG-Q[21–23] in the ON motor-state despite optimal medical management as verified by neurologist, stable PD medications at least one month prior to surgery and not expected to need changes during participation, able to walk independently for twenty metres without aid or rest, absence of secondary causes of gait problems, able to understand study requirements and provide consent, and finally, above 50 years of age. Exclusion criteria included presence of other significant neurological/psychiatric disorder or disease, presence of cognitive impairment or score of <23 on the MoCA, spinal abnormalities precluding SCS surgery, history of stroke or structural lesions on CT/MRI, history of chronic back pain (for RCT participants only), history of drug addiction/dependency, previous DBS surgery or infusion treatments for PD, pregnancy, and breast-feeding.

No changes to medication were allowed during the RCT.

The [18F]-FEOBV PET and T1 MR images of nine non-gait-impaired patients with idiopathic PD were included for comparison with our baseline images. These controls were recruited at our facility for another project and had no episodes of FoG or more than 1 point on the 3.10: Gait item of the MDS-UPDRS Part III while off-medication[42].

Clinical examinations

Clinical examinations were performed at the Department of Neurology, Aarhus University Hospital, Denmark, and were performed by two physicians (VSH and MHT), blinded to study allocation. Examinations were performed prior to surgery and at six- and twelve-month follow-ups while patients were in the ON motor state on their normal medication.

Patients were evaluated using the MDS-UPDRS[18], the Berg Balance Scale (BBS)[43], the NFOG-Q, and the Activities Balance Confidence Scale (ABC)[44]. Cognitive function was assessed using the MoCA. Quality of life was subjectively reported using the 39-item PDQ-39[20] and 36-item Short Form (SF-36)[45]. Furthermore, the mobility subscore of the PDQ-39 is reported separately. This subscore comprises the first ten questions of the scale (1: Difficulty doing leisure activities; 2: Difficulty looking after home; 3: Difficulty carrying bags of shopping; 4: Problems walking one kilometre; 5: Problems walking 100 metres; 6: Problems getting around the house; 7: Difficulty getting around in public; 8: Needed to be accompanied when out; 9: Frightened or worried about falling in public; 10: Been confined to the house more than liked). Each item is scored on a 5-point Likert scale: "never", "occasionally", "sometimes", "often", and "always", rated 0-4, 4 being most severe.

Gait characteristics were assessed by the three-metre TUG, TUG with cognitive and manual DT and timed 20-metre walks at comfortable speed and maximum speed, respectively. Finally, two 2-metre figure-of-eight walk tests were performed, one in each turning direction. All gait assessments were video recorded.

Patients' gait was additionally assessed using a home-worn triaxial accelerometer (STAT-ON, Sense4Care, Barcelona, Spain)[19] for home detection of FoG and falls prior to each study visit. Details on the use of the home-worn device are available in the Supplementary Materials.

All clinical and demographic data were entered into a Research Electronic Data Capture (REDcap) database managed by Aarhus University[46,47].

PET imaging acquisition and analysis

All participants were scanned at the Department of Nuclear Medicine and PET Centre, Aarhus University Hospital, Denmark. PET scans were performed at baseline and follow-up visits using a Siemens Biograph Vision 600 PET/CT system (Siemens/CTI, Knoxville, TN). In short, a standardised injection of 180 MBq (±10%) [18F]-FDG was administered, and images were acquired as static images 40 min post-injection. For [18F]-FDG imaging, patients were fasting for food for at least six hours and withheld PD medication for at least twelve hours before scans. For [18F]-FEOBV imaging, an injection of 200 MBq (± 10%) [18F]-FEOBV was administered, and images were acquired as static images in six bins of 300 seconds after a 180 min interval. Scans were performed on normal medication. MRI T1 imaging for co-registration was performed only at baseline using a 3.0 Tesla General Electric Company Signa PET/MRI for STEP-PD participants and a 3.0 Tesla Siemens Magnetom Skyra for controls.

PET images with standardised uptake values (SUV) were calculated using PMOD 4.0 (Bruker, Switzerland):

$$\mathrm{SUV} = \frac{\mathrm{imageactivityconcentation}}{\mathrm{decaycorrecteddosis}} * \mathrm{bodyweight} * 1000 \quad (1)$$

After co-registration to their individual SUV PET images, the MRIs were segmented and spatially normalised to the common Montreal Neurology Institute (MNI) space using the CAT12 tool of Statistical Parametric Mapping (SPM12, Functional Imaging Laboratory, UCL

Queen Square Institute of Neurology, London, UK) software in MATLAB R2024a (Mathworks Inc.) with DARTEL algorithm[48]. Subsequently, the SUV PET images were normalised by applying the obtained deformation fields in a final voxel size of 2 × 2 × 2 mm.

The spatially normalised SUV [^{18}F]-FDG PET images were then smoothed with a 4 × 4 × 4 Gaussian kernel. Using the Image Calculator in SPM12 software in MATLAB R2024a, SUVR [^{18}F]-FEOBV PET images were obtained by dividing each image (i1) by the reference SUV region in the middle of the centrum semiovale (CSO)[49] in each hemisphere:

$$\text{SUVRPETimage} = \frac{\text{SUVPETimage}}{\text{IndividualCSOSUV}} \quad (2)$$

Finally, SUVR [^{18}F]-FEOBV PET images were smoothed with a 6x6x6 Gaussian kernel.

For comparison of the RCT groups after six months of participation, we adjusted for baseline images by subtracting them from their individual six-month image. Within- and between-group comparisons were made with paired and two-group *t*-tests, respectively. The generated statistical maps were visualised using thresholds of 50 voxels and statistical alpha <0.05, <0.01 and <0.001. Results with family-wise error-corrected *p*-values < 0.05 were considered significant.

Outcome measures

The primary clinical outcome measure was the PIGD[15] subscore of the MDS-UPDRS, combining self-reported gait disorder and objective, clinician-based assessment of gait function. The PIGD score, ranging from 0–20 (20 being most severe), is calculated as the sum of patient reported items 12 and 13 of MDS-UPDRS Part II (2.12: Walking and balance, 2.13: Freezing of gait) and investigator assessed items 10–12 of MDS-UPDRS Part III (3.10: Gait, 3.11: Freezing of gait and 3.12: Postural stability). Items are scored 0–4, in ascending order of severity.

Furthermore, to achieve a better and objective characterisation of lower-body and gait disability we composed an additional MDS-UPDRS Part III subscore (Lower Body and Gait score, LBG score), which comprises the lower-body and gait subsections of the MDS-UPDRS part III (3.3 d&e: Rigidity, lower extremities; 3.7a&b: Toe tapping R/L; 3.8a&b: Leg agility R/L; 3.10: Gait; 3.11: Freezing of gait; and 3.12: Postural stability), ranging from 0–36 (36 being most severe).

Finally, to understand any effect on leg rigidity and bradykinesia of SCS, these items were summarised and presented independently.

All additional clinical scales (NFOG-Q, ABC, PDQ-39, PDQ-39 mobility subscore, BBS, SF-36, and Patient Global Impression of Change questionnaire) and all gait assessments described above as well as home-worn device derived measures (freezing episodes per 1000 steps, freezing duration, and falls per 100 steps) described in Supplementary Materials served as exploratory measures to inform designing of future trials.

To parallel our outcome measures (PIGD, LBG, bradykinesia and rigidity items, and MDS-UPDRS total motor score) in a natural cohort, we compared our cohort to similar patients from the PPMI[16]. PPMI patients with follow-up visits at both year seven and eight were included if they had H&Y stage II-III, data in ON motor-state, objective gait impairment and a history of FoG at year seven.

Safety

All adverse events were recorded. An adverse event (AE) was defined as a serious AE when it resulted in additional need for hospitalisation, including contact with emergency services or when it prolonged hospital stay by at least one day.

Surgical Implantation of SCS device

Implantation was performed according to standard practice at the Department of Neurosurgery, Aarhus University Hospital, Denmark. A 70 cm 8-contact wide-spaced percutaneous SCS lead (Boston Scientific, Marlborough, US) was placed in the epidural space through a percutaneous lumbar approach under fluoroscopic guidance by an experienced implanter (JCS, ANG, KM). Leads were advanced in the rostral direction until the lead was located at vertebral level T7-T10. Intraoperative placement testing was performed by a SCS specialist nurse (ALK). A conventional square-wave stimulation in the frequency range 40–100 Hz with adjustable pulse width (300–500 µs) was used for identifying the correct location. Optimal lead placement was defined as patient reporting paraesthesia in the lower back and both lower extremities, and adjustments were made to lead placement as needed to achieve this. The system was switched off and lead fixed to the spinal erector fascia and connected to a subcutaneously IPG (WaveWriter Alpha-Prime 16, Boston Scientific). To ensure optimal patient blinding, a non-rechargeable device was selected.

Randomisation and blinding

After completion of surgery with effective placement of the lead, all patients had two weeks of recovery with no stimulation before group allocation. This trial used the burst stimulation paradigm defined by the lead/IPG manufacturer, MicroBurst, which consists of a 40 Hz baseline followed by six pulses of 450 Hz. On the first day of treatment, the two pilot participants were programmed to MicroBurst as described below. For the remaining participants, randomisation was achieved by labels indicating "MicroBurst" or "Sham" distributed in identical envelopes in a 1:1 ratio. Labels were drawn at random, and participants were randomised consecutively according to the resultant label. The randomisation process was performed by an unblinded consultant neurosurgeon and an unblinded SCS specialist nurse. Programming of the IPG was attained out of view of the patient. Firstly, confirmation of lead placement was achieved by suprathreshold square-wave stimulation and adjustments to stimulation settings were performed as necessary based on patient-reported sensations. In the supine position, stimulation amplitude was progressively increased until paraesthesias were elicited in the lower extremities. Following paraesthesias elicitation, IPGs were either set to sham stimulation—IPG off—or to MicroBurst. The stimulation amplitude for MicroBurst was set at 50% of the paraesthesia limit on burst stimulation to ensure optimal patient blinding. Individual stimulation parameters are detailed in Supplementary Tables 14, 15.

At three months, correct placement was ensured in all participants by temporarily switching to square-wave stimulation and eliciting paraesthesias. Upon confirmation, patients were set again to either sham or MicroBurst as specified by their randomisation. At six months, the same procedure was performed before the patients were invited to enter the open-phase extension, where all patients were set to MicroBurst regardless of prior randomisation. Investigators were in regular contact with all participants, and if they did not experience any subjective effects after nine months of trial participation, they were invited to re-programming, changing the stimulation to square-wave 40–60 Hz stimulation based on the best achievement of lower limb paraesthesias for the remaining duration of the open phase. This protocol deviation was deemed necessary to prevent premature withdrawal.

Blinded examinators remained blinded during the whole duration of the RCT and statistical analyses of the RCT clinical assessments. Participants who entered the RCT remained blinded to their initial group location throughout the entire one-year study period. No muscle contractions were observed or reported by patients during programming visits or stimulation.

Statistical analysis

Between-group differences of sham vs MicroBurst at baseline were evaluated using Student's *t*-test. Assumptions of normal distribution and equal variance were verified by inspection of QQ-plots and the F-test. Between-group differences of six-month RCT-follow-up clinical

scores were estimated using a multiple linear regression model, entering baseline values as an adjusting measure. The model was checked by diagnostic plots of residuals. Estimates are given as end-of-treatment value with 95% confidence interval for the MicroBurst group with the sham group as reference.

Within-group differences comparing the change from pre-stimulation assessment to twelve-month follow-up were performed using paired *t*-tests. Post hoc analyses of participants with impactful improvement as well as six-month and one-year active treatment, respectively, were likewise evaluated using paired t-tests. The individual pre-stimulation assessment served as a baseline for these comparisons, either at trial time zero or six months, depending on original group allocation. Assumptions of normal distribution and equal variance were verified by inspection of QQ-plots, and the estimated differences are presented with 95% confidence intervals. All statistical tests were two-sided ($\alpha = 0.05$).

For outcomes with missing data, the number of individuals is specified in the outcome columns of the tables.

Medication changes

All participants were evaluated to be optimised in Parkinson medication prior to inclusion, and no participants made any changes to their medication during the RCT phase. However, during the open extension phase, some changes in medication were necessary due to non-gait-related Parkinsonian symptoms and medication side effects. No changes were made due to gait difficulties. One patient from the sham group attempted withdrawal of pramipexole due to behavioural side effects and was started on entacapone to maintain the daily levodopa equivalent dose. However, the patient did not manage to wean off the dopamine agonist but opted to stay on entacapone as well. This patient improved with one point in the PIGD score, but the bradykinesia and rigidity items did not change. One of the early starters had an increase from three to five in daily administrations of 125 mg levodopa/benserazide (one extra dose in the morning and evening for OFF periods). This patient had had the greatest improvement in gait assessments during the RCT phase prior to these medication changes. A second early starter had an increase in daily dose of levodopa equivalent dose of 187.50 mg and improved by three points in bradykinesia and rigidity items, but worsened by four points in PIGD score. Finally, one of the original pilot participants had to lower the pramipexole dose due to developing Pisa syndrome during the open extension phase, and this patient did score two points worse in PIGD score afterwards. Exclusion of these four patients with medication changes did not change any of the analyses presented above.

One consideration should be done in regard to the protocol deviation related to changes in dopamine replacement therapy in four patients during the open extension phase. While this might have had some effect on the patients' symptoms, it is unlikely that it had a major impact on our outcome measures. In fact, the exclusion of these four patients with medication changes did not change any of the results of the clinical analyses described in the paper.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The imaging data are available under restricted access for anonymity; access can be obtained by requests directed to the corresponding author, who will review requests on a case-by-case basis in accordance with Danish legal and ethical regulations. Response can be expected within two weeks. All source data and clinical, individual-level data generated in this study are provided in the Supplementary Materials/Source Data files. Source data are provided with this paper.

References

1. Bohnen, N. I. et al. Extra-nigral pathological conditions are common in Parkinson's disease with freezing of gait: an in vivo positron emission tomography study. *Mov. Disord.* **29**, 1118–1124 (2014).
2. Hvingelby, V. S. et al. Interventions to improve gait in Parkinson's disease: a systematic review of randomized controlled trials and network meta-analysis. *J. Neurol.* **269**, 4068–4079 (2022).
3. Bloem, B. R., Hausdorff, J. M., Visser, J. E. & Giladi, N. Falls and freezing of gait in Parkinson's disease: a review of two interconnected, episodic phenomena. *Mov. Disord.* **19**, 871–884 (2004).
4. Meier, K. et al. Examining the Duration Of Carryover Effect In Patients With Chronic Pain Treated With Spinal Cord Stimulation (EChO Study): an open, interventional, investigator-initiated, international multicenter study. *Neuromodulation* **27**, 887–898 (2024).
5. Agari, T. & Date, I. Spinal cord stimulation for the treatment of abnormal posture and gait disorder in patients with Parkinson's disease. *Neurol. Med Chir.* **52**, 470–474 (2012).
6. Fénelon, G. et al. Spinal cord stimulation for chronic pain improved motor function in a patient with Parkinson's disease. *Parkinsonism Relat. Disord.* **18**, 213–214 (2012).
7. Lin, Z. et al. High cervical spinal cord stimulation in Parkinson's disease with dopamine-resistant axial disabilities: a case with 2-year follow-up. *J. Neurol.* **270**, 3650–3653 (2023).
8. Zhou, P. B. & Bao, M. Spinal cord stimulation treatment for freezing of gait in Parkinson's disease: a case report. *Brain Stimul.* **15**, 76–77 (2022).
9. Hvingelby, V. S. et al. A pragmatic review on spinal cord stimulation therapy for parkinson's disease gait-related disorders: gaps and controversies. *Mov. Disord. Clin. Pract.* **11**, 927–947 (2024).
10. Ciocca, M., Seemungal, B. M. & Tai, Y. F. Spinal cord stimulation for gait disorders in Parkinson's disease and atypical parkinsonism: a systematic review of preclinical and clinical data. *Neuromodulation* **26**, 1339–1361 (2023).
11. Carra, R. B. et al. Spinal cord stimulation failed to improve parkinson's disease symptoms in randomized crossover double-blinded evaluation. *Mov. Disord.* **40**, 1279–1290 (2025).
12. Kobayashi, R. et al. New mode of burst spinal cord stimulation improved mental status as well as motor function in a patient with Parkinson's disease. *Parkinsonism Relat. Disord.* **57**, 82–83 (2018).
13. Chakravarthy, K. V. et al. Single-arm prospective multicenter case series on the use of burst stimulation to improve pain and motor symptoms in Parkinson's disease. *Bioelectron. Med* **6**, 18 (2020).
14. Hvingelby, V. S. et al. Spinal cord stimulation therapy for patients with Parkinson's disease and gait problems (STEP-PD): study protocol for an exploratory, double-blind, randomised, placebo-controlled feasibility trial. *BMJ Neurol. Open* **4**, e000333 (2022).
15. Stebbins, G. T. et al. How to identify tremor-dominant and postural instability/gait difficulty groups with the Movement Disorder Society Unified Parkinson's disease rating scale: Comparison with the Unified Parkinson's disease rating scale. *Mov. Disord.* **28**, 668–670 (2013).
16. The Parkinson Progression Marker Initiative (PPMI). *Prog. Neurobiol.* **95**, 629-635 (2011).
17. Postuma, R. B. et al. MDS clinical diagnostic criteria for Parkinson's disease. *Mov. Disord.* **30**, 1591–1601 (2015).
18. Goetz, C. G. et al. Movement Disorder Society-sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results. *Mov. Disord.* **23**, 2129–2170 (2008).
19. Rodríguez-Martín, D. et al. A new paradigm in Parkinson's disease evaluation with wearable medical devices: a review of STAT-ON™. *Front. Neurol.* **13**, 912343 (2022).
20. Peto, V., Jenkinson, C., Fitzpatrick, R. & Greenhall, R. The development and validation of a short measure of functioning and well-

being for individuals with Parkinson's disease. *Qual. Life Res* **4**, 241–248 (1995).
21. Nieuwboer, A. et al. Reliability of the new freezing of gait questionnaire: agreement between patients with Parkinson's disease and their carers. *Gait Posture* **30**, 459–463 (2009).
22. Hvingelby, V. S. et al. Prevalence and severity of freezing of gait in a Danish cohort of people with Parkinsonas disease. *Dan. Med. J.* **70**, A03230209 (2023).
23. Giladi, N. et al. Construction of the freezing of gait questionnaire for patients with Parkinsonism. *Parkinsonism Relat. Disord.* **6**, 165–170 (2000).
24. Nasreddine, Z. S. et al. The Montreal Cognitive Assessment, MoCA: a brief screening tool for mild cognitive impairment. *J. Am. Geriatr. Soc.* **53**, 695–699 (2005).
25. Boakye, M. et al. Spinal cord epidural stimulation for motor and autonomic function recovery after chronic spinal cord injury: a case series and technical note. *Surg. Neurol. Int* **14**, 87 (2023).
26. Ardolino, G. et al. Spinal direct current stimulation (tsDCS) in hereditary spastic paraplegias (HSP): a sham-controlled crossover study. *J. Spinal Cord. Med* **44**, 46–53 (2021).
27. Goodwin, B. J. et al. The efficacy of spinal cord stimulators in the reduction of multiple sclerosis spasticity: a narrative systematic review. *Brain Neurorehabil* **16**, e19 (2023).
28. Mayorova, L. et al. Immediate effects of anti-spastic epidural cervical spinal cord stimulation on functional connectivity of the central motor system in patients with stroke- and traumatic brain injury-induced spasticity: a pilot resting-state functional magnetic resonance imaging study. *Biomedicines* **11**, 2266 (2023).
29. Zhang, F. et al. Clinical features and related factors of freezing of gait in patients with Parkinson's disease. *Brain Behav.* **11**, e2359 (2021).
30. Luo, G. et al. Effects of STN-DBS surgery on cerebral glucose metabolism and distribution of DAT in Parkinson's disease. *Brain Behav.* **13**, e3172 (2023).
31. Poston, K. L. & Eidelberg, D. Functional brain networks and abnormal connectivity in the movement disorders. *Neuroimage* **62**, 2261–2270 (2012).
32. Meles, S. K. et al. Abnormal pattern of brain glucose metabolism in Parkinson's disease: replication in three European cohorts. *Eur. J. Nucl. Med Mol. Imaging* **47**, 437–450 (2020).
33. Fuentes, R., Petersson, P., Siesser, W. B., Caron, M. G. & Nicolelis, M. A. Spinal cord stimulation restores locomotion in animal models of Parkinson's disease. *Science* **323**, 1578–1582 (2009).
34. Yadav, A. P. & Nicolelis, M. A. L. Electrical stimulation of the dorsal columns of the spinal cord for Parkinson's disease. *Mov. Disord.* **32**, 820–832 (2017).
35. Takakusaki, K. Functional neuroanatomy for posture and gait control. *J. Mov. Disord.* **10**, 1–17 (2017).
36. Hanakawa, T. et al. Mechanisms underlying gait disturbance in Parkinson's disease: a single photon emission computed tomography study. *Brain* **122**, 1271–1282 (1999).
37. Augustine, J. R. Circuitry and functional aspects of the insular lobe in primates, including humans. *Brain Res Brain Res Rev.* **22**, 229–244 (1996).
38. de Lima-Pardini, A. C. et al. Brain networks associated with anticipatory postural adjustments in Parkinson's disease patients with freezing of gait. *Neuroimage Clin.* **28**, 102461 (2020).
39. Hassan, S., Amer, S., Alwaki, A. & Elborno, A. A patient with Parkinson's disease benefits from spinal cord stimulation. *J. Clin. Neurosci.* **20**, 1155–1156 (2013).
40. Nicolelis M. A., Fuentes R., Petersson P., Thevathasan W., Brown P. Spinal cord stimulation failed to relieve akinesia or restore locomotion in Parkinson's disease. *Neurology*. **75**, 1484 (2010).
41. Samotus, O., Parrent, A. & Jog, M. Long-term update of the effect of spinal cord stimulation in advanced Parkinson's disease patients. *Brain Stimul.* **13**, 1196–1197 (2020).
42. Horsager, J. et al. Mapping cholinergic synaptic loss in Parkinson's disease: an [18F]FEOBV PET case-control study. *J. Parkinsons Dis.* **12**, 2493–2506 (2022).
43. Berg, K. O., Wood-Dauphinee, S. L., Williams, J. I. & Maki, B. Measuring balance in the elderly: validation of an instrument. *Can. J. Public Health* **83**, S7–S11 (1992).
44. Powell, L. E. & Myers, A. M. The Activities-specific Balance Confidence (ABC) scale. *J. Gerontology: Ser. A* **50A**, M28–M34 (1995).
45. Ware, J. E. Jr. & Sherbourne, C. D. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. *Med Care* **30**, 473–483 (1992).
46. Harris, P. A. et al. The REDCap consortium: building an international community of software platform partners. *J. Biomed. Inform.* **95**, 103208 (2019).
47. Harris, P. A. et al. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *J. Biomed. Inform.* **42**, 377–381 (2009). 2009.
48. Ashburner, J. A fast diffeomorphic image registration algorithm. *NeuroImage (Orlando, Fla)* **38**, 95–113 (2007).
49. Aghourian, M. et al. Quantification of brain cholinergic denervation in Alzheimer's disease using PET imaging with [(18)F]-FEOBV. *Mol. Psychiatry* **22**, 1531–1538 (2017).

Acknowledgements

We thank all study participants for their valuable contributions and the personal at the Department of Nuclear Medicine, Aarhus University Hospital, for their skilled assistance throughout the study. This study was funded by the Danish Council for Independent Research (0134-00305B, NP). Funders were not involved in study design, data collection, analysis or manuscript writing.

Author contributions

M.H.T.: Data Curation, Formal Analysis, Investigation, Methodology, Project Administration, Visualisation, Writing—Original Draft Preparation. V.S.H.: Data Curation, Formal Analysis, Investigation, Methodology, Project Administration, Writing—Original Draft Preparation. Shared first authorship. The authorship order among the co–first authors was determined based on their specific contributions to the final manuscript writing. Both authors contributed equally to the overall study. E.L.J.: Investigation, Supervision, Writing – Review & Editing. M.M.: Investigation, Supervision, Writing – Review & Editing. E.H.D.: Investigation, Funding Acquisition, Writing – Review & Editing. T.H.: Conceptualisation, Funding Acquisition, Writing – Review & Editing. A.N.G.: Investigation, Supervision, Writing – Review & Editing. Y.T.: Conceptualisation, Funding Acquisition, Methodology, Supervision, Writing – Review & Editing. A.M.B.: Formal Analysis, Writing – Review & Editing. A.L.K.: Investigation, Writing – Review & Editing. R.N.V.: Project Administration, Writing – Review & Editing. J.H.: Investigation, Writing – Review & Editing. N.O.: Methodology, Writing—Review & Editing. A.S.M.A.: Project Administration, Writing—Review & Editing. K.M.: Investigation, Writing—Review & Editing. P.B.: Methodology, Writing—Review & Editing. S.M.: Conceptualisation, Funding Acquisition, Writing—Review & Editing. D.N.: Conceptualisation, Funding Acquisition, Writing—Review & Editing. E.M.: Conceptualisation, Supervision, Writing—Review & Editing. J.C.H.S.: Conceptualisation, Funding Acquisition, Investigation, Methodology, Supervision, Writing—Review & Editing. N.P.: Conceptualisation, Funding Acquisition, Methodology, Supervision, Writing – Original Draft Preparation

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41467-026-68782-w.

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Peer review information *Nature Communications* thanks Eduardo Moraud, Edoardo de Natale, and Yanzhong Wang for their contribution to the peer review of this work. A peer review file is available.

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